

Local Decoder.

David Ponarovsky

December 29, 2025

Introduction

Today:

- ▶ Proof.

1. Local Decoders.

Our Code

Code

In this session, we consider LDPC codes, where each bit adjoins to Δ_1 checks, and each check adjoins to Δ_2 bits. We denote by $\Delta = \Delta_1 \Delta_2$.

Local Decoder

definition

We say that D is a local decoder if when applied against $2p(1-p)$ -depolarized channel:

- ▶ There exists function $M : (0, 1) \rightarrow (0, 1)$ such that the probability of each bit being flipped is $M(2p(1-p)) < p^{\frac{100}{99}}$.
- ▶ There is a constant l such that for any bit, there are only $(\Delta)^l$ bits on which the event of it being flipped depends.

For example, if we take k large enough and consider the (n, k) -Toric, then both the swift-rule and the flip upon majority are local decoders.

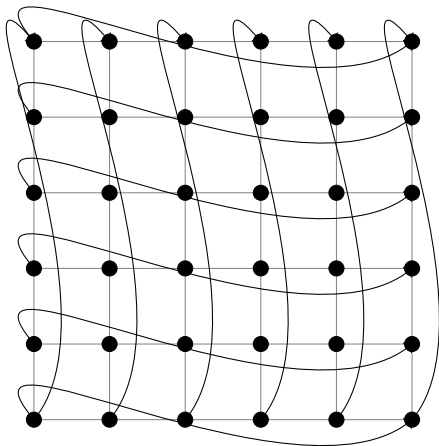


Figure: On the left is the Toric Graph. On the right are cross and face checks.

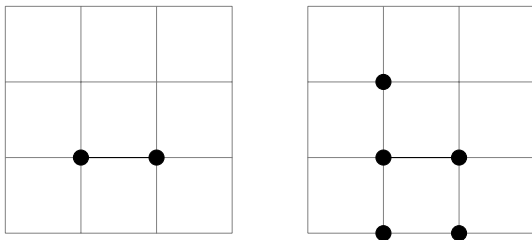


Figure: On the left is the Toric Graph. On the right are cross and face checks.

If both checks are unsatisfied, then flip the bit. Denote by κ the number of edges adjacent to those checks. Then the probability of failure is less than:

$$\begin{aligned}(1-p)2^{\kappa-1} \cdot p^2 + p2^{\kappa-1}p &\leq p^2 2^\kappa \\ &\leq p^{\frac{100}{99}}\end{aligned}$$

For p small enough.

Also correct for the local stochastic noise model. $\leq Cp^2 2^\kappa$.

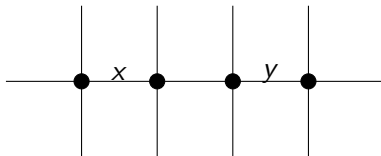


Figure: On the left is the Toric Graph. On the right are cross and face checks.

Denote by κ' the number of bits adjacent to a check that is adjacent to x but not to y ($\kappa' = \kappa - 1$). The probability that D fails to 'correct' both x and y , conditioned on no (q)bit in the intersection of their light cone being flipped, is less than:

$$\begin{aligned}
 & (1-p)^2 \left(2^{\kappa'-1} p^2\right)^2 + 2(1-p)p \left(2^{\kappa'-1} p \cdot 2^{\kappa'-1} p^2\right) \\
 & + p^2 \left(2^{\kappa'-1} p\right)^2 \\
 & \leq \left((1-p)2^{\kappa-1} \cdot p^2 + p2^{\kappa-1} p\right)^2 \\
 & \leq \left(p^{\frac{100}{99}}\right)^2
 \end{aligned}$$

In the local stochastic noise model the C is outside the power
 namely. $\leq C \left(p^{\frac{100}{99}}\right)^2$

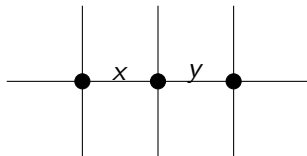


Figure: On the left is the Toric Graph. On the right are cross and face checks.

In the above case, where y is in the light cone of x and vice versa, the condition that there is no faulty bit on their light cone restricts to the case that both x and y weren't flipped, and no edge adjacent to the middle vertex is flipped. Thus, each of them sees at most one unsatisfied check. Thus, D doesn't flip them. Namely, the probability that D fails is zero $\leq p^{\frac{100}{99}}$.

Also correct for the local stochastic noise model.

Generalization

Consider a subset of qubits S . Suppose that for no two of them, all the qubits in the intersection of their light cones were flipped. Then the probability that D fails to 'correct' all the qubits in S is less than $p^{\frac{100}{99} \cdot |S|}$. ($Cp^{\frac{100}{99} \cdot |S|}$.)

Proof

If there are two qubits that share a check, then the decoder doesn't fail to 'correct' them, and therefore the probability that the decoder fails to 'correct' all the qubits in S is 0.

So, assume that the light cones of any two of them share only (q) bits. Thus, the probability is less than

$$\begin{aligned} & C \left((1-p)2^{\kappa'-1} \cdot p^2 + p2^{\kappa'-1}p \right)^{|S|} \\ & \leq C \left(p^{\frac{100}{99}} \right)^{|S|} \end{aligned}$$

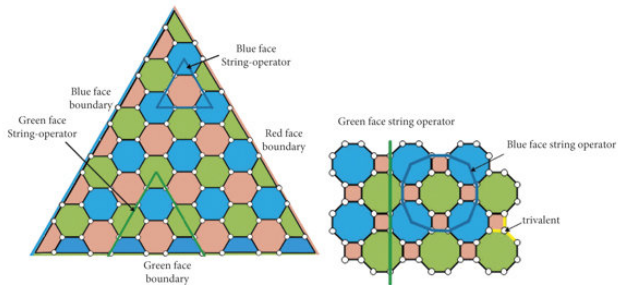


Figure: 2d color code

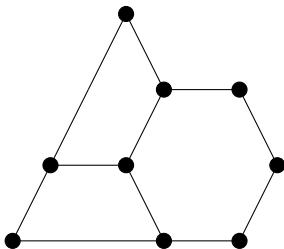


Figure: On the left is the Toric Graph. On the right are cross and face checks.

Boundary

Consider the qubit at the bottom left corner. The decoder will check all three checks and will flip the qubit if the bottom left check is the only one that is unsatisfied.

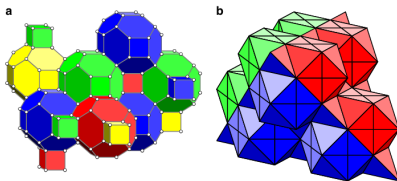


Figure 1 | Representations of the lattice geometry of the gauge color code. (a) In the primal picture qubits sit on the vertices of a four-valent lattice. The three-dimensional cells of the lattice are four colorable, that is, every cell can be assigned one of four colors such that it touches no other cell of the same color. (b) The gauge color code of linear size $L = 5$ drawn in the dual picture. Qubits lie on simplices of the lattice. The faces of the tetrahedra in the figure are given one of four colors such that no two faces of a given tetrahedron have the same color. The tetrahedra are stacked such that faces that touch always have the same color. For the gauge color code to encode a single logical qubit, the lattice must have four distinct, uniformly colored boundaries, as is shown in the figure. We give more details on the lattice construction in the dual picture in Supplementary Note 1 together with Supplementary Figs 1–6.

Figure: 3d color code

i++i